



Laboratory of Systems Tumor Immunology

Prof. Dr. Julio Vera-González

Telefon: 09131 85- 45876

Fax: 09131 85- 33701

E-Mail: julio.vera-gonzalez@uk-erlangen.de

www.jveralab.net

Hartmannstr. 14, 91052 Erlangen

Workplan for Student Project

Project title: DA3: Network-based analysis of melanoma therapy resistance

Supervisor: Julio Vera González, Prof.Dr. (Laboratory of Systems Tumorimmunology, Department of Dermatology, FAU Erlangen Nürnberg, www.jveralab.net)

Email: julio.vera-gonzalez@uk-erlangen.de

Aim

You got RNA-Seq data from melanoma patients that received immunotherapy. They have been classified in terms of their responsiveness to the therapy in three groups (Progressive Disease PD, Partial Response PR, and Complete Response CR). You also have access to a whole-genome network containing protein-protein and transcriptional regulation interactions between genes and proteins. Your aim is to generate a core network with genes that differentiate progressing disease patients (PD) from complete responders (CR).

Tasks

1. Prune the gene list by removing all genes that do not have expression above 1 TPM in at least 3 samples.
2. Next, obtain the network connecting only the genes that survived the filter in step 1. In this pruned network, compute the node degree (ND) and the betweenness centrality (BC) for all nodes that survived Step 1.
3. Compute the p-values and log2 FC for genes surviving Step 1. To this end, use the gene expression data in (normalized/raw) counts. Correct the significance of the p-values using Benjamini-Hochberg correction. Generate a volcano plot and perform a principal component analysis.
4. Score the nodes n_i utilizing the following score function:

$$S(ni) = 0.5 * \text{abs}(\log_2 FC(ni)) / \max(\text{abs}(\log_2 FC)) + 0.25 * ND(ni) / \max(ND) + 0.25 * BC(ni) / \max(BC)$$

5. Rank the genes in terms of the score and select the top 100 nodes.
6. Use the network from Step 2 to connect all top 100 nodes. Prune the unconnected single nodes. If the obtained network is smaller than 50 nodes, increase the size by selecting more nodes in Step 6.
7. Perform enrichment analysis for the nodes in the network.
8. Use information from existing databases to find drugs targeting the genes in the core network.

Dataset for Groups 1 and 2: Hugo et al. 2016. PubMed identifier PMID: 26997480

Dataset for Groups 3 and 4: Barrón-Gallardo et al. 2022 PubMed identifier PMID: 34981997

Relevant resources

Additional Datasets

You can look for papers containing datasets utilizing PubMed: <https://pubmed.ncbi.nlm.nih.gov/>

You can find datasets directly in GEO: <https://www.ncbi.nlm.nih.gov/geo/>

BioStudies: public repository of RNA Seq datasets <https://www.ebi.ac.uk/arrayexpress/>

Additional Networks

You can find melanoma networks in KEGG Pathway, or in www.vcells.net

Reactome: a database of curated pathways. <https://reactome.org/>

String: a database on protein-protein interactions useful to build networks. <https://string-db.org/>

DrugBank: A database of drug target interactions. <https://go.drugbank.com/>

Network visualization and analysis: Cytoscape

You can visualize networks and data-in-networks utilizing Cytoscape. This software also contains algorithms for core network selection (a.k.a. sub-network detection) as plugins.

You can perform differential expression analysis with DESeq2 and edgeR.

Most of the methods required can be utilized from python or Jupyter notebooks.

